

Question 1

Given:

$$P(A)=1/4, P(B)=1/2, P(C)=1/4$$

$$P(\text{Blue}|A)=4/12=1/3$$

$$P(\text{Blue}|B)=7/15$$

$$P(\text{Blue}|C)=2/15$$

Bayes Theorem:

$$P(C|\text{Blue}) = [P(\text{Blue}|C) \times P(C)] / \sum [P(\text{Blue}|B_i) \times P(B_i)]$$

$$= (2/15 \times 1/4) / [(1/3 \times 1/4) + (7/15 \times 1/2) + (2/15 \times 1/4)]$$

$$= (1/30) / (1/12 + 7/30 + 1/30)$$

$$= (1/30) / (7/20)$$

$$= 2/21 \approx 0.0952$$

Answer = 9.52% ($\approx 10\%$).

Question 2

$$P(A)=1/4, P(B)=1/2, P(C)=1/4$$

$$P(\text{Blue}|A)=3/10$$

$$P(\text{Blue}|B)=6/15=2/5$$

$$P(\text{Blue}|C)=2/15$$

$$P(C|\text{Blue}) = ((2/15)(1/4)) / [(3/10)(1/4) + (2/5)(1/2) + (2/15)(1/4)]$$

$$= (1/30) / (3/40 + 1/5 + 1/30)$$

$$= (1/30) / (37/120)$$

$$= 4/37 \approx 0.1081$$

Answer = 10.81%.

Question 3

Test samples/class = $1800 \times 20\% = 360$.

Actual positives = 360.

TP=210, TN=190, FP=30

FN = $360 - 210 = 150$

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \\ = (210 + 190) / 580 = 68.97\%$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) = 210 / 240 = 87.50\%$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) = 210 / 360 = 58.33\%$$

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP}) = 190 / 220 = 86.36\%$$

Inference: High precision but moderate recall, meaning several malignant cases are missed.

Question 4

Validation loss decreases until Epoch 3 then increases while training loss keeps decreasing.

- (i) Overfitting starts at Epoch 4.
- (ii) Techniques: Early stopping, Dropout/L2 regularization.
- (iii) Overfitting reduces generalization on unseen data.

Question 5

Validation accuracy peaks at Epoch 5 (73%) then decreases.

- (i) Overfitting starts after Epoch 5.
- (ii) Training accuracy increases because the model memorizes training data.
- (iii) Use Early Stopping and Data Augmentation/Dropout.

Question 6

Validation loss is minimum at Epoch 4.

- (i) Best generalization: Epoch 4.
- (ii) From Epoch 5 validation loss rises while training loss falls, indicating overfitting.
- (iii) L2 Regularization and Dropout improve generalization.

Question 7

$$\text{TP}=119, \text{TN}=120, \text{FP}=5, \text{FN}=6$$

$$\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total} = (119 + 120) / 250 = 95.60\%$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) = 119 / 124 = 95.97\%$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) = 119 / 125 = 95.20\%$$

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP}) = 120 / 125 = 96.00\%$$

$$F1=2\times\text{Precision}\times\text{Recall}/(\text{Precision}+\text{Recall})=95.58\%$$

Question 8

$$TP=138, TN=135, FP=15, FN=12$$

$$\text{Accuracy}=(TP+TN)/\text{Total}=(138+135)/300=91.00\%$$

$$\text{Precision}=TP/(TP+FP)=138/153=90.20\%$$

$$\text{Recall}=TP/(TP+FN)=138/150=92.00\%$$

$$\text{Specificity}=TN/(TN+FP)=135/150=90.00\%$$

$$F1=2\times\text{Precision}\times\text{Recall}/(\text{Precision}+\text{Recall})=91.09\%$$

Question 9

$$TP=180, TN=170, FP=30, FN=20$$

$$\text{Accuracy}=(TP+TN)/\text{Total}=(180+170)/400=87.50\%$$

$$\text{Precision}=TP/(TP+FP)=180/210=85.71\%$$

$$\text{Recall}=TP/(TP+FN)=180/200=90.00\%$$

$$\text{Specificity}=TN/(TN+FP)=170/200=85.00\%$$

$$F1=2\times\text{Precision}\times\text{Recall}/(\text{Precision}+\text{Recall})=87.80\%$$

Question 10

$$TP=230, TN=225, FP=25, FN=20$$

$$\text{Accuracy}=(TP+TN)/\text{Total}=(230+225)/500=91.00\%$$

$$\text{Precision}=TP/(TP+FP)=230/255=90.20\%$$

$$\text{Recall}=TP/(TP+FN)=230/250=92.00\%$$

$$\text{Specificity}=TN/(TN+FP)=225/250=90.00\%$$

$$F1=2\times\text{Precision}\times\text{Recall}/(\text{Precision}+\text{Recall})=91.09\%$$

Question 11

$$P(X)=1/3, P(Y)=2/3$$

$$P(A|X)=30/50=0.6$$

$$P(A|Y)=20/60=1/3$$

$$P(Y|A)=((1/3)(2/3))/[(0.6)(1/3)+(1/3)(2/3)]$$
$$=(2/9)/(19/45)=10/19=52.63\%$$

Question 12

$$P(A)=3/4, P(B)=1/4$$

$$P(D|A)=15/50=0.3$$

$$P(D|B)=25/50=0.5$$

$$P(B|D)=((0.5)(1/4))/[(0.3)(3/4)+(0.5)(1/4)]$$
$$=0.125/0.35=35.71\%$$

Question 13

Test samples/class=200.

$$FN=200-120=80$$

$$FP=200-150=50$$

$$\text{Accuracy}=(120+150)/400=67.5\%$$

$$\text{Precision}=120/170=70.59\%$$

$$\text{Recall}=120/200=60\%$$

$$\text{Specificity}=150/200=75\%$$

Question 14

Test samples/class=300.

$$FN=300-180=120$$

$$FP=300-160=140$$

$$\text{Accuracy}=(180+160)/600=56.67\%$$

$$\text{Precision}=180/320=56.25\%$$

$$\text{Recall}=180/300=60\%$$

$$\text{Specificity}=160/300=53.33\%$$

Question 15

Validation loss reaches its minimum (1.8) at Epoch 4.

(a) Best generalization: Epoch 4.

(b) Overfitting begins at Epoch 5 because validation loss increases while training loss continues decreasing.

(c) Two methods:

1. Early Stopping: Stops training before overfitting.
2. Dropout/L2 Regularization: Prevents memorization and improves generalization.